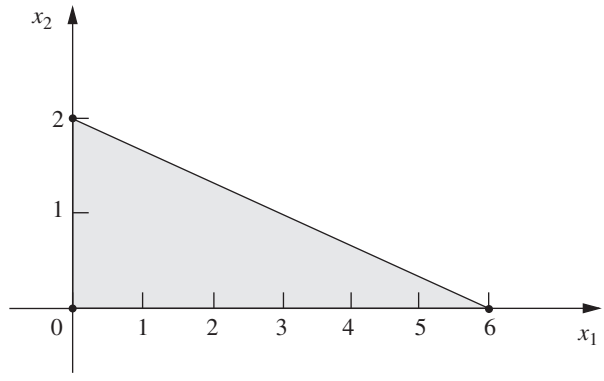
PARTIAL ANSWERS TO SELECTED PROBLEMS

CHAPTER 3

3.1-2. (a)



3.1-5. $(x_1, x_2) = (13, 5)$; $Z = 31$.

3.1-11. (b) $(x_1, x_2, x_3) = (26.19, 54.76, 20)$; $Z = 2{,}904.76$.

3.2-3. (b) Maximize $\quad Z = 9{,}000x_1 + 9{,}000x_2$,

subject to

$$\begin{aligned} x_1 \qquad\qquad &\leq 1 \\ x_2 &\leq 1 \\ 10{,}000x_1 + 8{,}000x_2 &\leq 12{,}000 \\ 400x_1 + 500x_2 &\leq 600 \end{aligned}$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0.$$

3.4-2. (a) *Proportionality*: OK since it is implied that a fixed fraction of the radiation dosage at a given entry point is absorbed by a given area.

Additivity: OK since it is stated that the radiation absorption from multiple beams is additive.

Divisibility: OK since beam strength can be any fractional level.

Certainty: Due to the complicated analysis required to estimate the data on radiation absorption in different tissue types, there is considerable uncertainty about the data, so sensitivity analysis should be used.

3.4-11. (b) From Factory 1, ship 200 units to Customer 2 and 200 units to Customer 3.
From Factory 2, ship 300 units to Customer 1 and 200 units to Customer 3.

3.4-12. (c) $Z = \$152{,}880$; $A_1 = 60{,}000$; $A_3 = 84{,}000$; $D_5 = 117{,}600$. All other decision variables are 0.

3.4-14. (b) Each optimal solution has $Z = \$13{,}330$.

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3.5-2. (c, e)

Resource	Resource Usage per Unit of Each Activity: Activity 1	Activity 2	Totals		Resource Available
1	2	1	10	≤	10
2	3	3	20	≤	20
3	2	4	20	≤	20
Unit Profit	20	30	$166.67		
Solution	3.333	3.333			

3.5-5. (a) Minimize $Z = 210C + 180T + 150A$,

subject to

$$90C + 20T + 40A \geq 200$$
$$30C + 80T + 60A \geq 180$$
$$10C + 20T + 60A \geq 150$$

and

$$C \geq 0, \qquad T \geq 0, \qquad A \geq 0.$$

CHAPTER 4

4.1-4. (a) The corner-point solutions that are *feasible* are (0, 0), (0, 1), $(\frac{1}{4}, 1)$, $(\frac{2}{3}, \frac{2}{3})$, $(1, \frac{1}{4})$, and (1, 0).

4.3-4. $(x_1, x_2, x_3) = (0, 10, 6\frac{2}{3})$; $Z = 70$.

4.6-1. (a, c) $(x_1, x_2) = (2, 1)$; $Z = 7$.

4.6-3. (a, c, e) $(x_1, x_2, x_3) = (\frac{4}{5}, \frac{9}{5}, 0)$; $Z = 7$.

4.6-9. (a, b, d) $(x_1, x_2, x_3) = (0, 15, 15)$; $Z = 90$.
(c) For both the Big M method and the two-phase method, only the final tableau represents a feasible solution for the real problem.

4.6-13. (a, c) $(x_1, x_2) = (-\frac{8}{7}, \frac{18}{7})$; $Z = \frac{80}{7}$.

4.7-5. (a) $(x_1, x_2, x_3) = (0, 1, 3)$; $Z = 7$.
(b) $y_1^* = \frac{1}{2}$, $y_2^* = \frac{5}{2}$, $y_3^* = 0$. These are the marginal values of resources 1, 2, and 3, respectively.

CHAPTER 5

5.1-1. (a) $(x_1, x_2) = (2, 2)$ is optimal. Other CPF solutions are (0, 0), (3, 0), and (0, 3).

5.1-12. $(x_1, x_2, x_3) = (0, 15, 15)$ is optimal.

5.2-2. $(x_1, x_2, x_3, x_4, x_5) = (0, 5, 0, \frac{5}{2}, 0)$; $Z = 50$.

5.3-1. (a) Right side is $Z = 8$, $x_2 = 14$, $x_6 = 5$, $x_3 = 11$.
(b) $x_1 = 0$, $2x_1 - 2x_2 + 3x_3 = 5$, $x_1 + x_2 - x_3 = 3$.

CHAPTER 6

6.1-1. (a) Minimize $W = 15y_1 + 12y_2 + 45y_3$,

subject to

$$-y_1 + y_2 + 5y_3 \geq 10$$
$$2y_1 + y_2 + 3y_3 \geq 20$$

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and

$$y_1 \geq 0, \qquad y_2 \geq 0, \qquad y_3 \geq 0.$$

6.3-1. (c)

Complementary Basic Solutions

Primal Problem			Dual Problem	
Basic Solution	**Feasible?**	$Z = W$	**Feasible?**	**Basic Solution**
(0, 0, 20, 10)	Yes	0	No	(0, 0, −6, −8)
(4, 0, 0, 6)	Yes	24	No	$\left(1\frac{1}{5}, 0, 0, -5\frac{3}{5}\right)$
(0, 5, 10, 0)	Yes	40	No	(0, 4, −2, 0)
$\left(2\frac{1}{2}, 3\frac{3}{4}, 0, 0\right)$	Yes and optimal	45	Yes and optimal	$\left(\frac{1}{2}, 3\frac{1}{2}, 0, 0\right)$
(10, 0, −30, 0)	No	60	Yes	(0, 6, 0, 4)
(0, 10, 0, −10)	No	80	Yes	(4, 0, 14, 0)

6.3-7. (c) Basic variables are x_1 and x_2. The other variables are nonbasic.
(e) $x_1 + 3x_2 + 2x_3 + 3x_4 + x_5 = 6, \;\; 4x_1 + 6x_2 + 5x_3 + 7x_4 + x_5 = 15, \;\; x_3 = 0, \;\; x_4 = 0, \;\; x_5 = 0.$ Optimal CPF solution is $(x_1, x_2, x_3, x_4, x_5) = (\frac{3}{2}, \frac{3}{2}, 0, 0, 0)$.

6.4-3. Maximize $\quad W = 8y_1 + 6y_2,$

subject to

$$\begin{aligned} y_1 + 3y_2 &\leq 2 \\ 4y_1 + 2y_2 &\leq 3 \\ 2y_1 \qquad &\leq 1 \end{aligned}$$

and

$$y_1 \geq 0, \qquad y_2 \geq 0.$$

6.4-8. (a) Minimize $\quad W = 120y_1 + 80y_2 + 100y_3,$

subject to

$$\begin{aligned} y_2 - 3y_3 &= -1 \\ 3y_1 - y_2 + y_3 &= 2 \\ y_1 - 4y_2 + 2y_3 &= 1 \end{aligned}$$

and

$$y_1 \geq 0, \qquad y_2 \geq 0, \qquad y_3 \geq 0.$$

CHAPTER 7

7.1-1. (d) Not optimal, since $2y_1 + 3y_2 \geq 3$ is violated for $y_1^* = \frac{1}{5}$, $y_2^* = \frac{3}{5}$.
(f) Not optimal, since $3y_1 + 2y_2 \geq 2$ is violated for $y_1^* = \frac{1}{5}$, $y_2^* = \frac{3}{5}$.

7.2-2.

Part	New Basic Solution $(x_1, x_2, x_3, x_4, x_5)$	Feasible?	Optimal?
(*a*)	(0, 30, 0, 0, −30)	No	No
(*b*)	(0, 20, 0, 0, −10)	No	No
(*c*)	(0, 10, 0, 0, 60)	Yes	Yes
(*d*)	(0, 20, 0, 0, 10)	Yes	Yes
(*e*)	(0, 20, 0, 0, 10)	Yes	Yes
(*f*)	(0, 10, 0, 0, 40)	Yes	No
(*g*)	(0, 20, 0, 0, 10)	Yes	Yes
(*h*)	(0, 20, 0, 0, 10, $x_6 = -10$)	No	No
(*i*)	(0, 20, 0, 0, 0)	Yes	Yes

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7.2-3. $-10 \le \theta \le \frac{10}{9}$

7.2-12. **(a)** $b_1 \ge 2$, $6 \le b_2 \le 18$, $12 \le b_3 \le 24$
(b) $0 \le c_1 \le \frac{15}{2}$, $c_2 \ge 2$

7.3-4. **(f)** The allowable range for the unit profit from producing toys is \$2.50 to \$5.00. The corresponding range for producing subassemblies is −\$3.00 to −\$1.50.

7.3-6. **(f)** For part (*a*), the change is within the allowable increase of \$10, so the optimal solution does not change. For part (*b*), the change is outside the allowable decrease of \$5, so the optimal solution might change. For part (*c*), the sum of the percentages of the allowable changes is 250 percent, so the 100 percent rule for simultaneous changes in objective function coefficients indicates that the optimal solution might change.

CHAPTER 8

8.1-2. $(x_1, x_2, x_3) = (\frac{2}{3}, 2, 0)$ with $Z = \frac{22}{3}$ is optimal.

8.1-6. **(a)** The new optimal solution is $(x_1, x_2, x_3, x_4, x_5) = (0, 0, 9, 3, 0)$ with $Z = 117$.

8.2-1. **(a, b)**

Range of θ	Optimal Solution	$Z(\theta)$
$0 \le \theta \le 2$	$(x_1, x_2) = (0, 5)$	$120 - 10\theta$
$2 \le \theta \le 8$	$(x_1, x_2) = \left(\frac{10}{3}, \frac{10}{3}\right)$	$\frac{320 - 10\theta}{3}$
$8 \le \theta$	$(x_1, x_2) = (5, 0)$	$40 + 5\theta$

8.2-4.

	Optimal Solution		
Range of θ	x_1	x_2	$Z(\theta)$
$0 \le \theta \le 1$	$10 + 2\theta$	$10 + 2\theta$	$30 + 6\theta$
$1 \le \theta \le 5$	$10 + 2\theta$	$15 - 3\theta$	$35 + \theta$
$5 \le \theta \le 25$	$25 - \theta$	0	$50 - 2\theta$

8.3-2. $(x_1, x_2, x_3) = (1, 3, 1)$ with $Z = 8$ is optimal.

CHAPTER 9

9.1-3. **(b)**

		Destination			
		Today	Tomorrow	Dummy	Supply
Source	Dick	3.0	2.7	0	5
	Harry	2.9	2.8	0	4
	Demand	3	4	2	

9.2-2. **(a)** Basic variables: $x_{11} = 4$, $x_{12} = 0$, $x_{22} = 4$, $x_{23} = 2$, $x_{24} = 0$, $x_{34} = 5$, $x_{35} = 1$, $x_{45} = 0$; $Z = 53$.
(b) Basic variables: $x_{11} = 4$, $x_{23} = 2$, $x_{25} = 4$, $x_{31} = 0$, $x_{32} = 0$, $x_{34} = 5$, $x_{35} = 1$, $x_{42} = 4$; $Z = 45$.
(c) Basic variables: $x_{11} = 4$, $x_{23} = 2$, $x_{25} = 4$, $x_{32} = 0$, $x_{34} = 5$, $x_{35} = 1$, $x_{41} = 0$, $x_{42} = 4$; $Z = 45$.

9.2-7. **(a)** $x_{11} = 3$, $x_{12} = 2$, $x_{22} = 1$, $x_{23} = 1$, $x_{33} = 1$, $x_{34} = 2$; three iterations to reach optimality.
(b, c) $x_{11} = 3$, $x_{12} = 0$, $x_{13} = 0$, $x_{14} = 2$, $x_{23} = 2$, $x_{32} = 3$; already optimal.

9.2-10. $x_{11} = 10$, $x_{12} = 15$, $x_{22} = 0$, $x_{23} = 5$, $x_{25} = 30$, $x_{33} = 20$, $x_{34} = 10$, $x_{44} = 10$; cost = \$77.30. Also have other tied optimal solutions.

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9.2-11. (b) Let x_{ij} be the shipment from plant i to distribution center j. Then $x_{13} = 2$, $x_{14} = 10$, $x_{22} = 9$, $x_{23} = 8$, $x_{31} = 10$, $x_{32} = 1$; cost = \$20,200.

9.3-4. (a)

		Task				
		Backstroke	**Breaststroke**	**Butterfly**	**Freestyle**	**Dummy**
	Carl	37.7	43.4	33.3	29.2	0
	Chris	32.9	33.1	28.5	26.4	0
Assignee	David	33.8	42.2	38.9	29.6	0
	Tony	37.0	34.7	30.4	28.5	0
	Ken	35.4	41.8	33.6	31.1	0

CHAPTER 10

10.3-4. (a) $O \to A \to B \to D \to T$ or $O \to A \to B \to E \to D \to T$, with length = 16.

10.4-1. (a) $\{(O, A); (A, B); (B, C); (B, E); (E, D); (D, T)\}$, with length = 18.

10.5-1.

Arc	(1, 2)	(1, 3)	(1, 4)	(2, 5)	(3, 4)	(3, 5)	(3, 6)	(4, 6)	(5, 7)	(6, 7)
Flow	4	4	1	4	1	0	3	2	4	5

10.8-3. (a) Critical path: Start $\to A \to C \to E \to$ Finish
Total duration = 12 weeks

(b) New plan:

Activity	**Duration**	**Cost**
A	3 weeks	\$54,000
B	3 weeks	65,000
C	3 weeks	68,666
D	2 weeks	41,500
E	2 weeks	80,000

\$7,834 is saved by this crashing schedule.

CHAPTER 11

11.3-2.

	Store		
	1	**2**	**3**
Allocations	1 3	2 2	2 0

11.3-7. (a)

Phase	**(*a*)**	**(*b*)**
1	2*M*	2.945*M*
2	1*M*	1.055*M*
3	1*M*	0
Market share	6%	6.302%

11.3-11. $x_1 = -2 + \sqrt{13} \approx 1.6056$, $x_2 = 5 - \sqrt{13} \approx 1.3944$; $Z = 98.233$.

11.4-3. Produce 2 on first production run; if none acceptable, produce 3 on second run. Expected cost = \$573.

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CHAPTER 12

12.1-2. (a) Minimize $Z = 4.5x_{em} + 7.8x_{ec} + 3.6x_{ed} + 2.9x_{el} + 4.9x_{sm} + 7.2x_{sc} + 4.3x_{sd} + 3.1x_{sl}$,

subject to

$$\begin{aligned} x_{em} + x_{ec} + x_{ed} + x_{el} &= 2 \\ x_{sm} + x_{sc} + x_{sd} + x_{sl} &= 2 \\ x_{em} + x_{sm} &= 1 \\ x_{ec} + x_{sc} &= 1 \\ x_{ed} + x_{sd} &= 1 \\ x_{el} + x_{sl} &= 1 \end{aligned}$$

and

all x_{ij} are binary.

12.3-1. (b)

Constraint	Product 1	Product 2	Product 3	Product 4	Totals		Modified Right-Hand Side	Original Right-Hand Side
First	5	3	6	4	6000	≤	6000	6000
Second	4	6	3	5	12000	≤	105999	6000
Marginal revenue	$70	$60	$90	$80	$80000			
Solution	0	2000	0	0				
	≤	≤	≤	≤				
	0	9999	0	0				
Set Up?	0	1	0	0	1	≤	2	
Start-up Cost	$50,000	$40,000	$70,000	$60,000				

Contingency Constraints:

Product 3:	0	≤	1	:Product 1 or 2
Product 4:	0	≤	1	:Product 1 or 2

Which Constraint (0 = First, 1 = Second):	0

12.3-5. (b, d) (long, medium, short) = (14, 0, 16), with profit of $95.6 million.

12.4-3. (b)

Constraint	Product 1	Product 2	Product 3	Total		Right-Hand Side
Milling	9	3	5	498	≤	500
Lathe	5	4	0	349	≤	350
Grinder	3	0	2	135	≤	150
Sales Potential	0	0	1	0	≤	20
Unit Profit	50	20	25	$2870		
Solution	45	31	0			
	≤	≤	≤			
	999	999	0			
Produce?	1	1	0	2	≤	2

12.4-5. (a) Let $x_{ij} = \begin{cases} 1 & \text{if arc } i \to j \text{ is included in shortest path} \\ 0 & \text{otherwise.} \end{cases}$

Mutually exclusive alternatives: For each column of arcs, exactly one arc is included in the shortest path. Contingent decisions: The shortest path leaves node i only if it enters node i.

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12.5-2. (a) $(x_1, x_2) = (2, 3)$ is optimal.
(b) None of the feasible rounded solutions are optimal for the integer programming problem.

12.6-1. $(x_1, x_2, x_3, x_4, x_5) = (0, 0, 1, 1, 1)$, with $Z = 6$.

12.6-7. (b)

Task	1	2	3	4	5
Assignee	1	3	2	4	5

12.6-9. $(x_1, x_2, x_3, x_4) = (0, 1, 1, 0)$, with $Z = 36$.

12.7-2. (a, b) $(x_1, x_2) = (2, 1)$ is optimal.

12.8-1. (a) $x_1 = 0, x_3 = 0$

CHAPTER 13

13.2-7. (a) Concave.

13.4-1. (a) Approximate solution = 1.0125.

13.5-3. Exact solution is $(x_1, x_2) = (2, -2)$.

13.5-7. (a) Approximate solution is $(x_1, x_2) = (0.75, 1.5)$.

13.6-3.

$$\begin{aligned}
-4x_1^3 - 4x_1 - 2x_2 + 2u_1 + u_2 &= 0 \quad (\text{or} \le 0 \text{ if } x_1 = 0).\\
-2x_1 - 8x_2 + u_1 + 2u_2 &= 0 \quad (\text{or} \le 0 \text{ if } x_2 = 0).\\
-2x_1 - x_2 + 10 &= 0 \quad (\text{or} \le 0 \text{ if } u_1 = 0).\\
-x_1 - 2x_2 + 10 &= 0 \quad (\text{or} \le 0 \text{ if } u_2 = 0).
\end{aligned}$$

$$x_1 \ge 0, \quad x_2 \ge 0, \quad u_1 \ge 0, \quad u_2 \ge 0.$$

13.6-6. $(x_1, x_2) = (1, 2)$ cannot be optimal.

13.6-8. (a) $(x_1, x_2) = (1 - 3^{-1/2}, 3^{-1/2})$.

13.7-2. (a) $(x_1, x_2) = (2, 0)$ is optimal.
(b) Minimize $\quad Z = z_1 + z_2,$

subject to

$$\begin{aligned}
2x_1 + u_1 - y_1 + z_1 &= 8\\
2x_2 + u_1 - y_2 + z_2 &= 4\\
x_1 + x_2 + v_1 &= 2
\end{aligned}$$

$$x_1 \ge 0, \quad x_2 \ge 0, \quad u_1 \ge 0, \quad y_1 \ge 0, \quad y_2 \ge 0, \quad v_1 \ge 0, \quad z_1 \ge 0, \quad z_2 \ge 0.$$

13.8-2. (b) Maximize $\quad Z = 3x_{11} - 3x_{12} - 15x_{13} + 4x_{21} - 4x_{23},$

subject to

$$\begin{aligned}
x_{11} + x_{12} + x_{13} + 3x_{21} + 3x_{22} + 3x_{23} &\le 8\\
5x_{11} + 5x_{12} + 5x_{13} + 2x_{21} + 2x_{22} + 2x_{23} &\le 14
\end{aligned}$$

and

$$0 \le x_{ij} \le 1, \quad \text{for } i = 1, 2, 3;\ j = 1, 2, 3.$$

13.9-8. (a) $(x_1, x_2) = \left(\frac{1}{3}, \frac{2}{3}\right)$.

13.9-14. (a) $P(x; r) = -2x_1 - (x_2 - 3)^2 - r\left(\frac{1}{x_1 - 3} + \frac{1}{x_2 - 3}\right)$.

(b) $(x_1, x_2) = \left[3 + \left(\frac{r}{2}\right)^{1/2}, 3 + \left(\frac{r}{2}\right)^{1/3}\right]$

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CHAPTER 14

14.2-2. The best solution found has links AC, BC, CD, and DE.

14.4-2. **(a)** For the first child, the options for the first link are 1-2, 1-8, 1-5, and 1-4 so the random numbers 0.09656 and 0.96657 say to choose link 1-2 and no mutation occurs. The options for the second link then are 2-3, 2-8, and 2-4, and so forth. A mutation occurs with the fifth link. The complete first child is 1-2-8-5-6-4-7-3-1.

CHAPTER 15

15.2-2. Player 1: strategy 2; player 2: strategy 1.

15.2-7. **(a)** Politician 1: issue 2; politician 2: issue 2.
(b) Politician 1: issue 1; politician 2: issue 2.

15.4-4. $(x_1, x_2) = (\frac{2}{5}, \frac{3}{5})$; $(y_1, y_2, y_3) = (\frac{1}{5}, 0, \frac{4}{5})$; $v = \frac{8}{5}$.

15.5-3. **(a)** Maximize x_4,

subject to

$$\begin{aligned} 5x_1 + 2x_2 + 3x_3 - x_4 &\geq 0 \\ 4x_2 + 2x_3 - x_4 &\geq 0 \\ 3x_1 + 3x_2 \qquad - x_4 &\geq 0 \\ x_1 + 2x_2 + 4x_3 - x_4 &\geq 0 \\ x_1 + x_2 + x_3 \qquad &= 1 \end{aligned}$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0, \qquad x_3 \geq 0, \qquad x_4 \geq 0.$$

CHAPTER 16

16.2-2. **(a)**

	State of Nature	
Alternative	**Sell 10,000**	**Sell 100,000**
Build Computers	0	54
Sell Rights	15	15

(c) Let p = prior probability of selling 10,000. They should build when $p \leq 0.722$, and sell when $p > 0.722$.

16.2-4. **(c)** Warren should make the countercyclical investment.

16.2-8. Order 25.

16.3-2. **(a)** EVPI = EP (with perfect info) − EP (without more info) = 34.5 − 27 = \$7.5 million.
(d)

Data:		*P* (Finding | State)	
		Finding	
State of Nature	**Prior Probability**	**Sell 10,000**	**Sell 100,000**
Sell 10,000	0.5	0.666666667	0.333333333
Sell 100,000	0.5	0.333333333	0.666666667

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Posterior Probabilities:		*P* (State | Finding) State of Nature	
Finding	***P* (Finding)**	**Sell 10,000**	**Sell 100,000**
Sell 10,000	0.5	0.666666667	0.333333333
Sell 100,000	0.5	0.333333333	0.666666667

16.3-4. **(b)** EVPI = EP (with perfect info) − EP (without more info) = 53 − 35 = \$18
(c) Betsy should consider spending up to \$18 to obtain more information.

16.3-8. **(a)** Up to \$230,000
(b) Order 25.

16.3-9. **(a)**

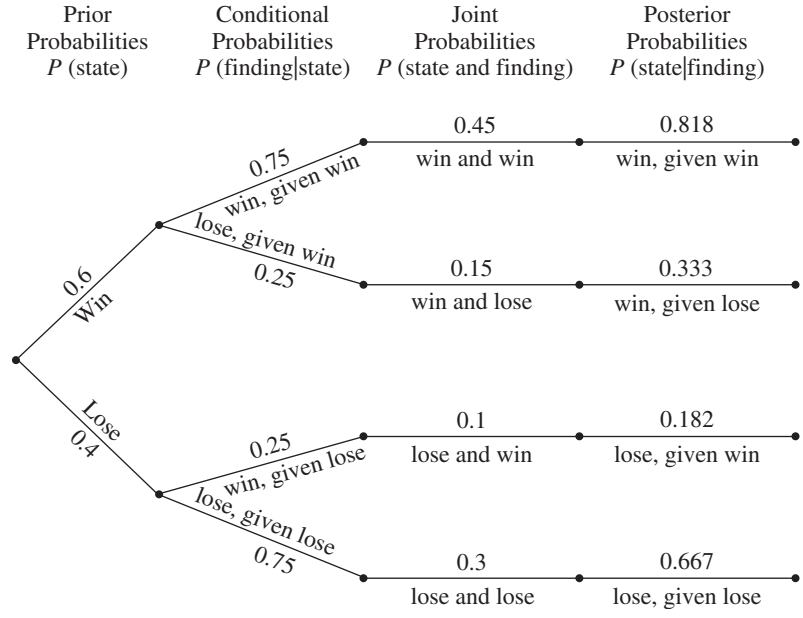
	State of Nature		
Alternative	**Poor Risk**	**Average Risk**	**Good Risk**
Extend Credit	−15,000	10,000	20,000
Don't Extend Credit	0	0	0
Prior Probabilities	0.2	0.5	0.3

(c) EVPI = EP (with perfect info) − EP (without more info) = 11,000 − 8,000 = \$3,000. This indicates that the credit-rating organization should not be used.

16.3-13. **(a)** Guess coin 1.
(b) Heads: coin 2; tails: coin 1.

16.4-2. The optimal policy is to do no market research and build the computers.

16.4-4. **(c)** EVPI = EP (with perfect info) − EP (without more info) = 1.8 − 1 = \$800,000
(d)



(f) Leland University should hire William. If he predicts a winning season then they should hold the campaign. If he predicts a losing season then they should not hold the campaign.

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16.5-7. **(a)** Choose to introduce the new product (expected payoff is \$12.5 million).
(b) \$7.5 million.
(c) The optimal policy is not to test but to introduce the new product.

16.6-2. **(a)** Choose not to buy insurance (expected payoff is \$249,840).
(b) $U(\text{insurance}) = 499.82$
$U(\text{no insurance}) = 499.8$
Optimal policy is to buy insurance.

16.6-4. $U(10) = 9$

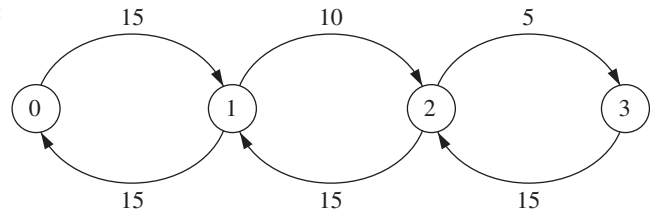
CHAPTER 17

17.2-1. Input source: population having hair; customers: customers needing haircuts; and so forth for the queue, queue discipline, and service mechanism.

17.2-2. **(b)** $L_q = 0.375$
(d) $W - W_q = 24.375$ minutes

17.4-2. **(c)** 0.0527

17.5-5. **(a)** State:



(c) $P_0 = \frac{9}{26}$, $P_1 = \frac{9}{26}$, $P_2 = \frac{3}{13}$, $P_3 = \frac{1}{13}$.
(d) $W = 0.11$ hour.

17.5-8. **(b)** $P_0 = \frac{2}{5}$, $P_n = (\frac{3}{5})(\frac{1}{2})^n$
(c) $L = \frac{6}{5}$, $L_q = \frac{3}{5}$, $W = \frac{1}{25}$, $W_q = \frac{1}{50}$

17.6-2. **(a)** $P_0 + P_1 + P_2 + P_3 + P_4 = 0.96875$ or 97 percent of the time.

17.6-21. **(a)** Combined expected waiting time = 0.211
(c) An expected process time of 3.43 minutes would cause the expected waiting times to be the same for the two procedures.

17.6-26. **(a)** 0.429

17.6-32. **(a)** three machines
(b) three operators

17.7-1. **(a)** W_q (exponential) $= 2W_q$ (constant) $= \frac{8}{5}W_q$ (Erlang).
(b) W_q (new) $= \frac{1}{2}W_q$ (old) and L_q (new) $= L_q$ (old) for all distributions.

17.7-6. **(a, b)** Under the current policy an airplane loses 1 day of flying time as opposed to 3.25 days under the proposed policy.
Under the current policy 1 airplane is losing flying time per day as opposed to 0.8125 airplane.

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17.7-9.

Service Distribution	P_0	P_1	P_2	L
Erlang	0.561	0.316	0.123	0.561
Exponential	0.571	0.286	0.143	0.571

17.8-1. **(a)** This system is an example of a nonpreemptive priority queueing system.

(c) $\dfrac{W_q \text{ for first-class passengers}}{W_q \text{ for coach-class passengers}} = \dfrac{0.033}{0.083} = 0.4$

17.8-4. **(a)** $W = \frac{1}{2}$
(b) $W_1 = 0.20$, $W_2 = 0.35$, $W_3 = 1.10$
(c) $W_1 = 0.125$, $W_2 = 0.3125$, $W_3 = 1.250$

17.10-2. 4 cash registers

CHAPTER 18

18.3-1. **(a)** $t = 1.83$, $Q = 54.77$
(b) $t = 1.91$, $Q = 57.45$, $S = 52.22$

18.3-3. **(a)**

Data

$d =$	676	(demand/year)
$K =$	\$75	(setup cost)
$h =$	\$600.00	(unit holding cost)
$L =$	3.5	(lead time in days)
$WD =$	365	(working days/year)

Results

Reorder point =	6.5
Annual setup cost =	\$10,140
Annual holding cost =	\$ 1,500
Total variable cost =	\$11,640

Decision

$Q =$	5	(order quantity)

(d)

Data

$d =$	676	(demand/year)
$K =$	\$75	(setup cost)
$h =$	\$600	(unit holding cost)
$L =$	3.5	(lead time in days)
$WD =$	365	(working days/year)

Results

Reorder point =	6.48
Annual setup cost =	\$3,900
Annual holding cost =	\$3,900
Total variable cost =	\$7,800

Decision

$Q =$	13	(order quantity)

The results are the same as those obtained in part (*c*).
(f) Number of orders per year = 52
$ROP = 6.5 -$ inventory level when each order is placed
(g) The optimal policy reduces the total variable inventory cost by \$3,840 per year, which is a 33 percent reduction.

18.3-6. **(a)** $h = \$3$ per month which is 15 percent of the acquisition cost.
(c) Reorder point is 10.
(d) $ROP = 5$ hammers, which adds \$20 to his TVC (5 hammers $\times$ \$4 holding cost).

18.3-7. $t = 3.26$, $Q = 26{,}046$, $S = 24{,}572$

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18.3-12. **(a)** Optimal $Q = 500$

18.4-4. Produce 3 units in period 1 and 4 units in period 3.

18.6-6. **(b)** Ground Chuck: $R = 145$.
Chuck Wagon: $R = 829$.
(c) Ground Chuck: safety stock = 45.
Chuck Wagon: safety stock = 329.
(f) Ground Chuck: \$39,378.71.
Chuck Wagon: \$41,958.61.
Jed should choose Ground Chuck as their supplier.
(g) If Jed would like to use the beef within a month of receiving it, then Ground Chuck is the better choice. The order quantity with Ground Chuck is roughly 1 month's supply, whereas with Chuck Wagon the optimal order quantity is roughly 3 month's supply.

18.7-5. **(a)** Optimal service level = 0.667
(c) $Q^* = 500$
(d) The probability of running short is 0.333.
(e) Optimal service level = 0.833

CHAPTER 19

19.2-2. **(c)** Use slow service when no customers or one customer is present and fast service when two customers are present.

19.2-3. **(a)** The possible states of the car are dented and not dented.
(c) When the car is not dented, park it on the street in one space. When the car is dented, get it repaired.

19.2-5. **(c)** State 0: attempt ace; state 1: attempt lob.

19.3-2. **(a)** Minimize $\quad Z = 4.5y_{02} + 5y_{03} + 50y_{14} + 9y_{15},$

subject to

$$y_{01} + y_{02} + y_{03} + y_{14} + y_{15} = 1$$
$$y_{01} + y_{02} + y_{03} - \left(\frac{9}{10}y_{01} + \frac{49}{50}y_{02} + y_{03} + y_{14}\right) = 0$$
$$y_{14} + y_{15} - \left(\frac{1}{10}y_{01} + \frac{1}{50}y_{02} + y_{15}\right) = 0$$

and

all $y_{ik} \geq 0$.

19.3-4. **(a)** Minimize $\quad Z = -\frac{1}{8}y_{01} + \frac{7}{24}y_{02} + \frac{1}{2}y_{11} + \frac{5}{12}y_{12},$

subject to

$$y_{01} + y_{02} - \left(\frac{3}{8}y_{01} + y_{11} + \frac{7}{8}y_{02} + y_{12}\right) = 0$$
$$y_{11} + y_{12} - \left(\frac{5}{8}y_{01} + \frac{1}{8}y_{02}\right) = 0$$
$$y_{01} + y_{02} + y_{11} + y_{12} = 1$$

and

$y_{ik} \geq 0 \qquad$ for $i = 0, 1;\ k = 1, 2.$

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CHAPTER 20

20.1-1. (b) Let the numbers 0.0000 to 0.5999 correspond to strikes and the numbers 0.6000 to 0.9999 correspond to balls. The random observations for pitches are 0.7520 = ball, 0.4184 = strike, 0.4189 = strike, 0.5982 = strike, 0.9559 = ball, and 0.1403 = strike.

20.1-10. (b) Use $\lambda = 4$ and $\mu = 5$.

(i) Answers will vary. The option of training the two current mechanics significantly decreases the waiting time for German cars, without a significant impact on the wait for Japanese cars, and does so without the added cost of a third mechanic. Adding a third mechanic lowers the average wait for German cars even more, but comes at an added cost for the third mechanic.

20.3-1. (a) 5, 8, 1, 4, 7, 0, 3, 6, 9, 2

20.4-2. (b) $F(x) = 0.0965$ when $x = -5.18$
$F(x) = 0.5692$ when $x = 18.46$
$F(x) = 0.6658$ when $x = 23.29$

20.4-6. (a) Here is a sample replication.

Summary of Results:

Win? (1 = Yes, 0 = No)	0
Number of Tosses =	3

Simulated Tosses

Toss	Die 1	Die 2	Sum
1	4	2	6
2	3	2	5
3	6	1	7
4	5	2	7
5	4	4	8
6	1	4	5
7	2	6	8

Results

Win?	Lose?	Continue?
0	0	Yes
0	0	Yes
0	1	No
NA	NA	No
NA	NA	No
NA	NA	No
NA	NA	No

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